

Question:

At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<think>

The brain’s overall developmental trajectory shows consistent, predictable patterns: neuronal migration and synaptic maturation progress from caudal to rostral regions, and myelination (an important determinant of speed and efficiency) similarly advances cranially. Specifically, the telencephalon (cortex and associated structures) matures more slowly than structures that arise earlier in ontogeny, such as the brain stem and cerebellum.

By birth, the brain stem (medulla, pons, midbrain) and cerebellum (medially, fast-spinning climbing fibers innervating deep cerebellar nuclei; broadly, cortical foliation and connections) show substantial myelination and synaptic refinement. The thalamus has largely finished migration and de novo synapses, and parietal cortex is relatively mature because it emerges early in cerebral development. By contrast, frontal cortex shows prominent GABAergic staining at term, reflecting its primary role in controlling myelination, and posterior cingulate/precuneus often forms only later in postnatal life. Despite macroscopic appearance at birth, the cortex itself remains the last major region to undergo extensive synaptic elaboration, myelination, and structural refinement; clinically, neonates and infants express heightened plasticity and maturational lag in cognitive and motor functions.

Therefore, the least developed part of the brain at birth is the cortex.

Correct answer: G. cortex

</think>

<answer>

G. cortex

</answer>

